POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Chemistry

Semester: Fall

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What do you mean by acidic buffer? Derive Henderson Hasselbalch equation for acidic buffer. Calculate the P^H of 100 ml of 0.4NH_3 solution in which 20ml. of 0.5M Hcl is added. (PK_b = 4.74) 8

   b) What do you mean by standard hydrogen electrode (SHE)? Calculate the emf of the cell at 25°C. 7

   $E^0_{Fe/Fe++}$ = + 0.44V $E^0_{Ag/Ag}$ = -0.80V

   [Fe++] = 0.5M, [Ag+] = 0.2m

2. a) Give reasons: 9

   i) Second ionization energy is greater than first ionization energy

   ii) Electron affinity of fluorine is less than chlorine

   iii) Ionization energy of gallium is higher than aluminium

   b) Explain why zinc sulphate salt is colorless whereas copper sulphate salt is blue colored. Transition metal compounds are generally paramagnetic in nature, explain it. 6

3. a) What is elimination reaction? Write the mechanism of E_1 and E_2 reaction with one proper example of each. 8

   b) Define free radical. Write the reaction mechanism of free radical addition reaction. 7

4. a) What is condensation polymerization? Write the preparation, properties and uses of Nylon – 6, 6. 7

   b) Write short note on: 8

   i) Processing of natural rubber

   ii) Silicones

5. a) What is the principle of TLC? Mention its applications in analytical field. 5

b) Define explosive. Give the methods of preparation of TNT and its important uses. 5

c) What are three important raw materials used in the manufacture of cement? Explain reaction mechanism of setting of cement. 5

6. a) What is soil pollution? Discuss its effects on agriculture and living beings. How can it be controlled? 7

b) What are particulates? Classify particulates, and discuss in brief its effects on human. How can use control global warming and particulates? 8

7. Write short notes on: **(Any two)** 2×5

a) Stereoisomerism

b) Ozone layer Depletion

c) Mass spectroscopy